Citations & Variables for Heatmaps

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Construal Level Theory (climate change experiments): variables and associated citations

Variables	Citation
• environmental engagement score • proenvironmental behaviour	Attanasi et al., 2025
• objective temporal distance • confidence in economic recovery • perceived sustainability objectives • proenvironmental intentions	Balaguer-Mercado A. et al., 2023
• psychological proximity • exposure to positive visions • environmental identity • eco-tech use intentions • proenvironmental intentions	Bosone et al., 2024
• risk framing vs efficacy • objective geographic distance • response efficacy perception • objective geographic distance (-) • proenvironmental policy support • risk perception • proenvironmental intentions	Chu and Yang, 2020
• self-efficacy perception • interactivity • contingency • proenvironmental behaviour • psychological distance framing (-) • proenvironmental policy support • risk perception	Fox et al., 2020

• proenvironmental behaviour • psychological distance (-)	Guillard et al., 2021
• gender (men) • peer contributions • proenvironmental value-action gap • perceived personal impact • immediacy (-) • uncertainty • age	Hoffmann et al., 2024
• proenvironmental intentions • nostalgic message • China • childhood memory recall • temporal psychological distance (-) • USA • nostalgia	Huang J. et al., 2024
• proenvironmental behaviour • desktop intervention • emotional intensity • negative affect • proenvironmental intentions • response efficacy • virtual reality • temporal psychological distance (-)	Hurrell et al., 2024
• low construal • adaptation framing • low construal thinking • high construal • mitigation framing • high construal thinking • proenvironmental behaviour • risk perception • construal message mismatch (-) • proenvironmental intentions	Jia Y. et al., 2021

• knowledge level • objective temporal distance • objective temporal distance (-) • perceived severity • proenvironmental intentions	Thomsen, 2024
• gender (women) • cost (-) • policy attributes • education • longer residence • income (-) • psychological distance (-) • proenvironmental policy support • objective temporal distance (-)	Wei Y. et al., 2023
• proenvironmental support • hope • imagination task (vs control)	Wenzel et al., 2024
• anticipated regret if carpooling (-) • psychological distance from pandemic (+) • anticipated regret if not carpooling • proenvironmental intentions	Xiao Q. et al., 2024
• proenvironmental behaviour • risk perception • virtual reality	van Gevelt T. et al., 2023

Table 1: Table of variables presented in the heatmap (CLT climate change experiments) and their associated citations.

Construal Level Theory (climate change surveys): variables and associated citations

Variables	Citation
• nature walks • geographic psychological distance (-) • temporal psychological distance (-) • social psychological distance (-) • proenvironmental intentions (limit dispoable) • proenvironmental intentions (buy regional) • proenvironmental intentions (eco purchase) • proenvironmental intentions (water conservation)	Bosone and Bertoldo, 2022
• gender (girls) • caring beliefs • dispositional empathy • threat perception (condor) • psychological distance (-) • proenvironmental intentions	Büscher M. et al., 2023
• psychological distance (-) • proenvironmental intentions	Carmi and Kimhi, 2015

• negative emotions • issue salience • flood experience • issue type • climate concern • proenvironmental intentions • proenvironmental policy support	Demski et al., 2017
• proximity to covid • subjective norms • extrinsic motives • perceived control • attitude • proenvironmental intentions	D'Arco M. et al., 2022
• flood exposure • drought exposure • place attachment • risk perception • psychological distance (-) • proenvironmental behaviour	Guillard M. et al., 2021
• global climate concern • perceived local water availability shortage • proenvironmental intentions	Haden et al., 2012

• climate change belief • age (65+) • political affiliation (Democrat) • proenvironmental behaviour • proenvironmental policy support	Hao et al., 2020
• personal norm (proenvironmental) • ascribed responsibility • proenvironmental intentions	Kim et al., 2022
• psychological distance • fear of covid19 • air pollution concern • proenvironmental behaviour	Liu et al., 2021
• distant desirability • subjective norms • attitude • attitude towards proenvironmental products • perceived behavioural control • near desirability • distant feasibility • proenvironmental intentions (eco purchase)	Shabnam et al., 2021

• perceived benefits of carbon capture and storage • positive affect • political affiliation (Democrat) • risk perception • psychological distance (-) • proenvironmental tech support	Shah P. et al., 2023
• certainty • impact on developing countries • objective temporal distance (-) • objective geographic distance (-) • proenvironmental intentions	Spence et al., 2011
• age • green self-identity • pro-eco worldview • habitual worry • anxious emotions • gender (women) • environmental values • proenvironmental behaviour	Verplanken et al., 2020

• doubtful (audience segment) • alarmed (audience segment) • cautious (audience segment) • self/response efficacy perception • social norms • concerned (audience segment) • objective temporal distance (-) • hypothetical distance (-) • proenvironmental intentions	Waldo J.L. et al., 2025
• education • age (36-40) • gender (women) • environmental values • risk perception • psychological distance (-) • proenvironmental behaviour	Wu M. et al., 2023

Table 2: Table of variables presented in the heatmap (CLT climate change surveys) and their associated citations.

Construal Level Theory (infectious disease control): variables and associated citations

Variables	Citation
• high-involvement investors • high-involvement investors (-) • positive SRI screenings • risk aversion • income • impact criteria • financial attitudes • psychological distance (-) • socially responsible intentions • public health	Apostolakis G. et al., 2016

• perceived severity (-)	Hmielowski J.D. et al., 2023
• conservative media exposure (-)	
• nonpartisan media use	
• psychological distance	
• perceived severity	
• psychological distance (-)	
• covid19 vaccine intentions	
• public health	
• temporal psychological distance	Johnson, 2018
• zika concern (-)	
• self-relevance	
• objective geographic distance	
• zika concern	
• mosquito control intentions	
• zika containment intentions	
• travel avoidance intentions	
• safe-sex intentions	
• public health	
• virtual reality	Xu Z. et al., 2024
• vaccination intentions	
• public health	

• virtual reality • augmented reality • video • social distancing intentions • self-test intentions • vaccination intentions • self-test behaviour • booster uptake • social distancing & mask wearing • public health	Xu and Dam, 2024
• loss framing • future-focused • objective temporal distance (-) • covid19 vaccine intentions • public health	Xu and Wang, 2025

• risk perception (-) • trait empathy • political affiliation (conservative) (-) • social psychological distance (-) • prosocial emotions • risk perception • monkeypox vaccination policy support • monkeypox vaccination intentions • public health	Yang and Dong, 2024

Table 3: Table of variables presented in the heatmap (CLT infectious disease control) and their associated citations.

Consideration of Future Consequences: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• future-focused environment • politically conservative (-) • women (-) • altruism • NGO familiarity • hope condition • environmental attitude • age • future-focused • climate intentions • climate behaviour • climate policy support	Lagomarsino and Lemarié, 2024
• temporal discounting • future-focused • climate behaviour	Macaskill A.C. et al., 2019

• eco-anxiety • cognitive alternatives • present-focused (-) • present-focused • future-focused • climate intentions • radical climate intentions	Pittaway et al., 2024
• future-generation framing • empathy induction (-) • honesty-humility (-) • future-focused (-) • recent antibiotic use • men • disease severity • antibiotic use (hypothetical)	Santana et al., 2023
• reputational concern • future-focused (condition) • health donation willingness • health volunteer willingness	Sjåstad, 2019

• temporal distance (-) • loss framing • future-focused • Covid19 vaccine intentions	Xu and Wang, 2025
• determinance • sustainable identity • biospheric values • relevance of sustainability attributes • determinance of sustainability attributes • future-focused • climate behaviour	van Dam and Fischer, 2015
• product category • relevance of sustainability attributes • determinance of sustainability attributes • future-focused • climate behaviour	van Dam and van Trijp, 2013

Table 4: Table of variables presented in the heatmap (Consideration of Future Consequences) and their associated citations.

Temporal Discounting: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• health stress • temporal discounting (-) • Covid19 containment measure compliance	Agrawal M. et al., 2023
• risk perception • risk perception (-) • optimism pandemic • depression symptoms • temporal discounting (-) • Covid19 containment measure compliance • Covid19 containment measure non-compliance	Calluso C. et al., 2021
• stress • temporal discounting • Covid19 containment measure non-compliance • stockpiling during Covid19	DeAngelis BN et al., 2022

• collective responsibility (-) • complacency • unemployment • temporal discounting • Covid19 vaccine hesitancy	Freitas-Lemos R. et al., 2023
• risk aversion • present-bias • preference for short-term gains vs biodiversity	Ihli et al., 2022
• objective temporal distance (-) • climate public goods contribution	Jacquet J. et al., 2013
• objective temporal distance (-) • social distance (-) • uncertainty (-) • climate action willingness	Kaplan B.A. et al., 2014
• objective temporal distance (-) • beneficiary type (future others vs present others) • moral obligation • prosocial donation willingness • prosocial policy support • prosocial donation	Law KF et al., 2024

• income (-) • sensitivity to reward magnitude • age (-) • temporal discounting • Covid19 containment measure non-compliance	Lloyd A et al., 2021
• age • internal locus of control • authority respect • longtermism • temporal discounting (-) • Covid19 containment measure compliance • Covid19 containment measure non-compliance	Lo Presti S. et al., 2022
• future-focused • temporal discounting • climate action	Macaskill A.C. et al., 2019
• gender (men) • risk aversion • fear of side effects (-) • subjective vaccine knowledge • objective vaccine knowledge • temporal discounting (-) • gender (women) • flu vaccination	Nuscheler R et al., 2016

• length of payment • biodiversity tourism vs seed supply program • familiarity with marine program • income • age • climate action willingness	Vasquez-Lavín et al., 2019
• psychological distance(-) • objective temporal distance (-) • education • income (-) • longer residence • psychological distance (-) • policy attributes • cost (-) • gender (women) • climate action willingness • climate policy support	Wei Y. et al., 2023

Table 5: Table of variables presented in the heatmap (Temporal Discounting) and their associated citations.

Anticipated Emotion: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• men • women • perceived benefits of vaccination • anticipated regret if unvaccinated • HPV vaccine intention	Christy et al., 2016
• graphical images vs text • anticipated regret questions condition • message involvement • HPV vaccination intention	Cox et al., 2014
• positive framing • negative framing • anticipated pride of purchase • eco-rating level • price level (-) • climate intention	Gorissen et al., 2024

• unknown-effectiveness condition • prevention focus • vaccine effectiveness condition • perceived vaccination necessity • perceived vaccine safety • anticipated regret if unvaccinated • flu vaccination uptake • flu vaccination intention	Leder et al., 2015
• perceived infection probability • anticipated worry if unvaccinated • past vaccination • age • anticipated regret if vaccinated (-) • flu vaccination uptake • flu vaccination intention	Liao et al., 2013
• past vaccination • anticipated counterfactural relief • eligibility for free vaccine • anticipated regret if unvaccinated • flu vaccination uptake • flu vaccination intention	Lorimer et al., 2024

• instrumental attitudes about vaccination • barriers to vaccination • subjective norms about vaccination • low political trust (-) • age • anticipated regret if unvaccinated • covid19 vaccination intention	Rountree and Prentice, 2022
• health profession (midwives) • past vaccination • decisional uncertainty • agreement with vaccination policy • moral responsibility to vaccinate • cost-benefit perceptions • vaccination beliefs • anticipated regret if unvaccinated • vaccination attitude • pertussis vaccination intention	Visser et al., 2016
• negative affect • anticipatory emotion condition • infection risk perception • covid19 prevention behaviours • mask wearing	Wang et al., 2022

• psychological distance from pandemic (+) • anticipated regret if not carpooling • anticipated regret if carpooling (-) • climate intention	Xiao et al., 2024
• perceived cancer likelihood • decision urgency • perceived cancer severity likelihood • anticipated regret if unvaccinated • anticipated regret if vaccinated (-) • HPV vaccination intention • HPV vaccination uptake	Ziarnowski et al., 2009

Table 6: Table of variables presented in the heatmap (Anticipated Emotion) and their associated citations.

Cognitive Alternatives: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• psychological distance (-) • environmental identity • exposure to postive eco future vision conditions • climate intentions • eco-tech use intentions	Bosone et al., 2024
• utopian propensity • utopian affect • environmental cognitive alternatives • climate intentions • climate behaviour • citizenship intentions	Chevrier et al., 2024

• utopian thinking • expansive altruism • longtermism • environmental cognitive alternatives • climate policy support • justice policy support • future generations policy support	Law et al., 2023
• eco-anxiety • future-focused thinking • present-focused (-) • present-focused • cognitive alternatives • climate intentions • radical climate intentions	Pittaway et al., 2024
• legacy concerns • longtermism • cognitive alternatives • climate behaviour	Stylianos Syropoulos et al., 2024

• government efficacy • government efficacy (-) • collective efficacy (-) • fear • counterarguing • cautionary message • positive message • environmental cognitive alternatives • climate intentions	Troy et al., 2024
• environmental cognitive alternatives • climate intentions • climate behaviour	Wright et al., 2020
• environmental identity • cognitive alternatives • climate intentions • climate behaviour	Wright et al., 2021

Table 7: Table of variables presented in the heatmap (Cognitive Alternatives) and their associated citations.

Episodic Future Thinking: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• EFT condition • proenvironmental behaviour	Bø and Wolff, 2020
• childhood memory recall • nostalgia • proenvironmental intentions • risk perception • temporal psychological distance (-)	Huang J. et al., 2024
• expectations (COVID-19 × vaccine) • expectations of severe COVID-19 outcomes (-) • negative COVID-19 fantasies • negative COVID-19 fantasies × vaccine fantasies • negative COVID-19 impact expectation • negative vaccine impact expectation • provaccine information search • vaccination status (vaccinated)	Kim S. et al., 2024

• vaccination uptake • vaccine fantasies • vaccine fantasies (−1 SD) • vaccine hesitancy	Kim S. et al., 2024
• expansive altruism • longtermism • objective temporal distance (-) • perceived positive impact • subjective visualisation ability	Law KF et al., 2024
• anticipatory emotion condition • covid19 prevention behaviours • infection risk perception • mask wearing • negative affect	Wang XTX et al., 2022
• hope • imagination task (vs control) • proenvironmental support	Wenzel et al., 2024

• SW China (-) • affective attitude • altruistic values • descriptive norms • disasters imagery (-) • education • emotional arousal • environmental attitude • experience imagery • future • gender (male) • human • impact imagery (-) • income • nature imagery • prescriptive norms • proenvironmental behaviour	Yangli et al., 2025

Table 8: Table of variables presented in the heatmap (Episodic Future Thinking) and their associated citations.

Utopian Thinking: significant predictors, mediators, moderators, and dependent variables

Variables	Citation
• hope • social justice intentions • system justification (-)	Badaan et al., 2022
• climate policy support • climate resilience • social justice • social justice policy support • social justice protest intentions • tuiph factor collective orienation • tuiph factor economic transformation • tuiph factor transformative motivation • tuiph general factor	Basso and Krpan, 2022

• climate intentions • eco-tech use intentions • environmental identity • exposure to positive visions • psychological proximity	Bosone et al., 2024
• antiutopian thinking (-) • citizenship intentions • climate behaviour • climate intentions • environmental cognitive alternatives • utopian affect • utopian propensity	Chevrier et al., 2024
• climate intentions • dystopian thinking • fear • hope • hope (-) • utopian thinking	Daysh et al., 2024

• antiutopian thinking (-) • citizenship engagement • citizenship intentions • future -> present sequence • system justification (-) • utopian thinking	Fernando et al., 2018
• climate behaviour • climate intentions • evaluation of depicted society • green vs scifi utopia • participative efficacy • social change motivation	Fernando et al., 2019
• climate policy support • environmental cognitive alternatives • expansive altruism • future generations policy support • justice policy support • longtermism • utopian thinking	Law et al., 2023

Table 9: Table of variables presented in the heatmap (Utopian Thinking) and their associated citations.